Case Study 5: Traffic Violation Penalty Calculator

Aim: To develop a Python program that calculates traffic violation penalties based on vehicle category, offence type, repeat-offence status, and predefined fine rules. The program also identifies conditions that may lead to driving licence disqualification.

1. Problem Statement

A traffic department needs a system to calculate fines for multiple traffic violations committed by vehicle drivers.

The program should:

1. Accept vehicle category.

2. Accept offence type.

3. Check whether the offence is repeated.

4. Apply predefined fine amounts.

5. Calculate the total penalty.

6. Identify serious violations.

7. Check driving licence disqualification conditions.

8. Generate a final violation report.

9. Assumed Fine Rules

The following amounts are illustrative values for this academic case study, not a statement of current legal penalties.

Offence	Fine
Over Speeding	₹1,000
No Helmet	₹500
No Seat Belt	₹500
Signal Jumping	₹1,000
Drunk Driving	₹5,000

Additional rules:

- Repeat offence: Double the original fine.
- Drunk driving: Disqualification review required.
- Three or more recorded violations: Disqualification review required.
- Vehicle categories: Two-wheeler, four-wheeler and heavy vehicle.

3. Algorithm

4. Start.
5. Enter vehicle category.
6. Enter the number of violations.
7. For each violation:
   - Enter offence type.
   - Check whether it is a repeat offence.
   - Identify the predefined fine.
   - Calculate the applicable penalty.
   - Add the penalty to the total fine.
   - Check disqualification conditions.
5. Display individual violation details.
6. Calculate the total penalty.
7. Display disqualification status.
8. Stop.

9. Pseudocode

```
START

INPUT vehicle_category
INPUT number_of_violations

SET total_fine = 0
SET violation_count = 0
```

```
SET disqualification = FALSE

FOR each violation

    INPUT offence_type
    INPUT repeat_status

    IF offence_type = "Over Speeding"
        fine = 1000

    ELSE IF offence_type = "No Helmet"
        fine = 500

    ELSE IF offence_type = "No Seat Belt"
        fine = 500

    ELSE IF offence_type = "Signal Jumping"
        fine = 1000

    ELSE IF offence_type = "Drunk Driving"
        fine = 5000
        disqualification = TRUE

    ELSE
        DISPLAY Invalid offence
        CONTINUE
```

```
    END IF

    IF repeat_status = YES
        fine = fine * 2
    END IF

    total_fine = total_fine + fine

    violation_count = violation_count + 1

END FOR

IF violation_count >= 3
    disqualification = TRUE
END IF

DISPLAY total_fine
DISPLAY disqualification status

STOP
```

5. Python Program

Complete Python solution

```
# Traffic Violation Penalty Calculator# Predefined fine rulesfine_rules = {    "over speeding": 1000,    "no helmet": 500,    "no seat belt": 500,    "signal jumping": 1000,    "drunk driving": 5000}# Vehicle categoryvehicle = input(    "Enter vehicle category (two-wheeler/four-wheeler/heavy vehicle): ").lower()# Validate vehicle categoryif vehicle not
```

```
in ["two-wheeler", "four-wheeler", "heavy vehicle"]:    print("Invalid vehicle category!")else:    n = int(input("Enter number of violations: "))    total_fine = 0 violation_count = 0    disqualification = False    violations = []    # Process multiple violations    for i in range(n):        print("\nViolation", i + 1)
```

6. Sample Output

Enter vehicle category: two-wheeler

Enter number of violations: 3

Violation 1

Enter offence type: no helmet

Is this a repeat offence? yes

Penalty: Rs. 1000

Violation 2

Enter offence type: signal jumping

Is this a repeat offence? no

Penalty: Rs. 1000

Violation 3

Enter offence type: over speeding

Is this a repeat offence? no

Penalty: Rs. 1000

====== TRAFFIC VIOLATION REPORT ======

Vehicle Category: two-wheeler

Offence Details:

Offence: no helmet

Repeat Offence: yes

Fine: Rs. 1000

Offence: signal jumping

Repeat Offence: no

Fine: Rs. 1000

Offence: over speeding

Repeat Offence: no

Fine: Rs. 1000

Total Violations: 3

Total Penalty: Rs. 3000

Disqualification Status: REVIEW REQUIRED

====== REPORT COMPLETED ======

7. Flowchart

#chatgpt-mermaid-_r_27c_{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;fill:rgb(237, 237, 237);}@keyframes edge-animation-frame{from{stroke-dashoffset:0;}}@keyframes dash{to{stroke-dashoffset:0;}}#chatgpt-mermaid-_r_27c_ .edge-animation-slow{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 50s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_27c_ .edge-animation-fast{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 20s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_27c_ .error-icon{fill:rgb(48, 48, 48);}#chatgpt-mermaid-_r_27c_ .error-text{fill:rgb(237, 237, 237);stroke:rgb(237, 237, 237);}#chatgpt-mermaid-_r_27c_ .edge-thickness-normal{stroke-width:1px;}#chatgpt-mermaid-_r_27c_ .edge-thickness-thick{stroke-width:3.5px;}#chatgpt-mermaid-_r_27c_ .edge-pattern-solid{stroke-dasharray:0;}#chatgpt-mermaid-_r_27c_ .edge-thickness-invisible{stroke-width:0;fill:none;}#chatgpt-mermaid-_r_27c_ .edge-pattern-dashed{stroke-dasharray:3;}#chatgpt-mermaid-_r_27c_ .edge-pattern-dotted{stroke-dasharray:2;}#chatgpt-mermaid-_r_27c_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_27c_ .marker.cross{stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_27c_ svg{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;}#chatgpt-mermaid-_r_27c_ p{margin:0;}#chatgpt-mermaid-_r_27c_ .label{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_27c_ .cluster-label text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_27c_ .cluster-label span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_27c_ .cluster-label span p{background-color:transparent;}#chatgpt-mermaid-_r_27c_ .label text,#chatgpt-mermaid-_r_27c_ span{fill:rgb(237, 237, 237);color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_27c_ .node rect,#chatgpt-mermaid-_r_27c_ .node circle,#chatgpt-mermaid-_r_27c_ .node ellipse,#chatgpt-mermaid-_r_27c_ .node polygon,#chatgpt-mermaid-_r_27c_ .node path{fill:rgb(9, 23, 44);stroke:rgb(31, 78, 148);stroke-width:1px;}#chatgpt-mermaid-_r_27c_ .rough-node .label text,#chatgpt-mermaid-_r_27c_ .node .label text,#chatgpt-mermaid-_r_27c_ .image-shape .label,#chatgpt-mermaid-_r_27c_ .icon-shape .label{text-anchor:middle;}#chatgpt-mermaid-_r_27c_ .node .katex path{fill:#000;stroke:#000;stroke-width:1px;}#chatgpt-mermaid-_r_27c_ .rough-node .label,#chatgpt-mermaid-_r_27c_ .node .label,#chatgpt-mermaid-_r_27c_ .image-shape .label,#chatgpt-mermaid-_r_27c_ .icon-shape .label{text-align:center;}#chatgpt-mermaid-_r_27c_ .node.clickable{cursor:pointer;}#chatgpt-mermaid-_r_27c_ .root .anchor path{fill:rgb(175, 175, 175)!important;stroke-width:0;stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_27c_ .arrowheadPath{fill:rgb(175, 175, 175);}#chatgpt-mermaid-_r_27c_ .edgePath .path{stroke:rgb(175, 175, 175);stroke-

width:1px;}#chatgpt-mermaid-_r_27c_ .flowchart-link{stroke:rgb(175, 175, 175);fill:none;}#chatgpt-mermaid-_r_27c_ .edgeLabel{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_27c_ .edgeLabel p{background-color:rgb(0, 0, 0);}#chatgpt-mermaid-_r_27c_ .edgeLabel rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_27c_ .labelBkg{background-color:rgba(0, 0, 0, 0.5);}#chatgpt-mermaid-_r_27c_ .cluster rect{fill:rgb(48, 48, 48);stroke:rgba(255, 255, 255, 0.15);stroke-width:1px;}#chatgpt-mermaid-_r_27c_ .cluster text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_27c_ .cluster span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_27c_ div.mermaidTooltip{position:absolute;text-align:center;max-width:200px;padding:2px;font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:12px;background:rgb(48, 48, 48);border:1px solid rgba(255, 255, 255, 0.15);border-radius:2px;pointer-events:none;z-index:100;}#chatgpt-mermaid-_r_27c_ .flowchartTitleText{text-anchor:middle;font-size:18px;fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_27c_ rect.text{fill:none;stroke-width:0;}#chatgpt-mermaid-_r_27c_ .icon-shape,#chatgpt-mermaid-_r_27c_ .image-shape{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_27c_ .icon-shape p,#chatgpt-mermaid-_r_27c_ .image-shape p{background-color:rgb(0, 0, 0);padding:2px;}#chatgpt-mermaid-_r_27c_ .icon-shape .label rect,#chatgpt-mermaid-_r_27c_ .image-shape .label rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_27c_ .label-icon{display:inline-block;height:1em;overflow:visible;vertical-align:-0.125em;}#chatgpt-mermaid-_r_27c_ .node .label-icon path{fill:currentColor;stroke:revert;stroke-width:revert;}#chatgpt-mermaid-_r_27c_ .node .neo-node{stroke:rgb(31, 78, 148);}#chatgpt-mermaid-_r_27c_ [data-look="neo"].node rect,#chatgpt-mermaid-_r_27c_ [data-look="neo"].cluster rect,#chatgpt-mermaid-_r_27c_ [data-look="neo"].node polygon{stroke:url(#chatgpt-mermaid-_r_27c_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_27c_ [data-look="neo"].swimlane.cluster rect{filter:none;}#chatgpt-mermaid-_r_27c_ [data-look="neo"].node path{stroke:url(#chatgpt-mermaid-_r_27c_-gradient);stroke-width:1px;}#chatgpt-mermaid-_r_27c_ [data-look="neo"].node .outer-path{filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_27c_ [data-look="neo"].node .neo-line path{stroke:rgb(31, 78, 148);filter:none;}#chatgpt-mermaid-_r_27c_ [data-look="neo"].node circle{stroke:url(#chatgpt-mermaid-_r_27c_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_27c_ [data-look="neo"].node circle .state-start{fill:#000000;}#chatgpt-mermaid-_r_27c_ [data-look="neo"].icon-shape .icon{fill:url(#chatgpt-mermaid-_r_27c_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_27c_ [data-look="neo"].icon-shape .icon-neo path{stroke:url(#chatgpt-mermaid-_r_27c_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_27c_ .node text{font-size:14px;font-weight:600;letter-

spacing:normal;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_27c_ .edgeLabels text{font-size:13px;font-weight:600;letter-spacing:-0.08px;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_27c_ .node tspan[font-weight="normal"],#chatgpt-mermaid-_r_27c_ .edgeLabels tspan[font-weight="normal"]{font-weight:600;}#chatgpt-mermaid-_r_27c_ .edgeLabel .label rect{opacity:1;rx:13px;ry:13px;fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-width:1px;}#chatgpt-mermaid-_r_27c_ .node rect,#chatgpt-mermaid-_r_27c_ .node circle,#chatgpt-mermaid-_r_27c_ .node ellipse,#chatgpt-mermaid-_r_27c_ .node polygon,#chatgpt-mermaid-_r_27c_ .node path{fill:rgb(0, 40, 77);stroke:rgba(255, 255, 255, 0.1);stroke-width:1px;}#chatgpt-mermaid-_r_27c_ .node rect{rx:16px;ry:16px;}#chatgpt-mermaid-_r_27c_ .mermaid-decision .label-container{fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-dasharray:2,2;}#chatgpt-mermaid-_r_27c_ .edgePaths .flowchart-link{stroke:rgb(175, 175, 175);stroke-width:1px;stroke-linecap:round;stroke-linejoin:round;}#chatgpt-mermaid-_r_27c_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_27c_ :root{--mermaid-font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";}StartEnter vehicle categoryEnter number of violationsEnter offence and repeat statusValid offence?Skip invalid violationFind predefined fineRepeat offence?Double the fineKeep original fineAdd fine to totalDrunk driving?Flag for disqualification reviewContinueMore violations?Three or more violations?Display final penalty and statusStopNoYesYesNoYesNoYesNoYesNo

8. Concepts Used from Syllabus

Syllabus concept	Application
Variables	Fine, total penalty
Data types	Integer and string
Operators	Addition, multiplication, comparison
Conditional statements	Fine calculation
Loops	Processing multiple violations
Strings	Offence names

Dictionaries	Predefined fine rules
Lists	Storing violation details
Tuples	Individual violation records
Tuple unpacking	Displaying violation records

9. Conclusion

The Traffic Violation Penalty Calculator automates the calculation of traffic fines based on offence type and repeat-offence status. It calculates the total penalty and identifies cases requiring disqualification review. This program demonstrates the practical application of Python programming concepts in traffic management.

Note: The fine amounts and disqualification rules are simplified assumptions for educational purposes. Actual penalties and licence-disqualification decisions depend on applicable traffic laws and official procedures.